

Watch-DevKit-HW

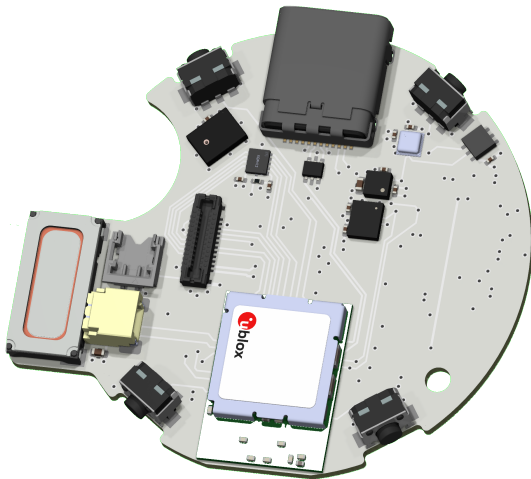
2026-02-21

Variant: CHECKED

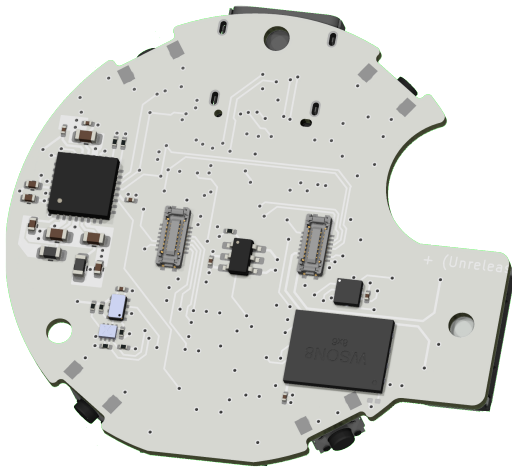
Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11		21		31	
2	Block Diagram	12		22		32	
3	Power - Topology	13		23		33	
4	Project Architecture	14		24		34	
5	Power Management	15		25		35	
6	Peripherals	16		26		36	
7	MCU	17		27		37	
8	Extension	18		28		38	
9	Revision History	19		29		39	
10		20		30		40	

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 21-Feb-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

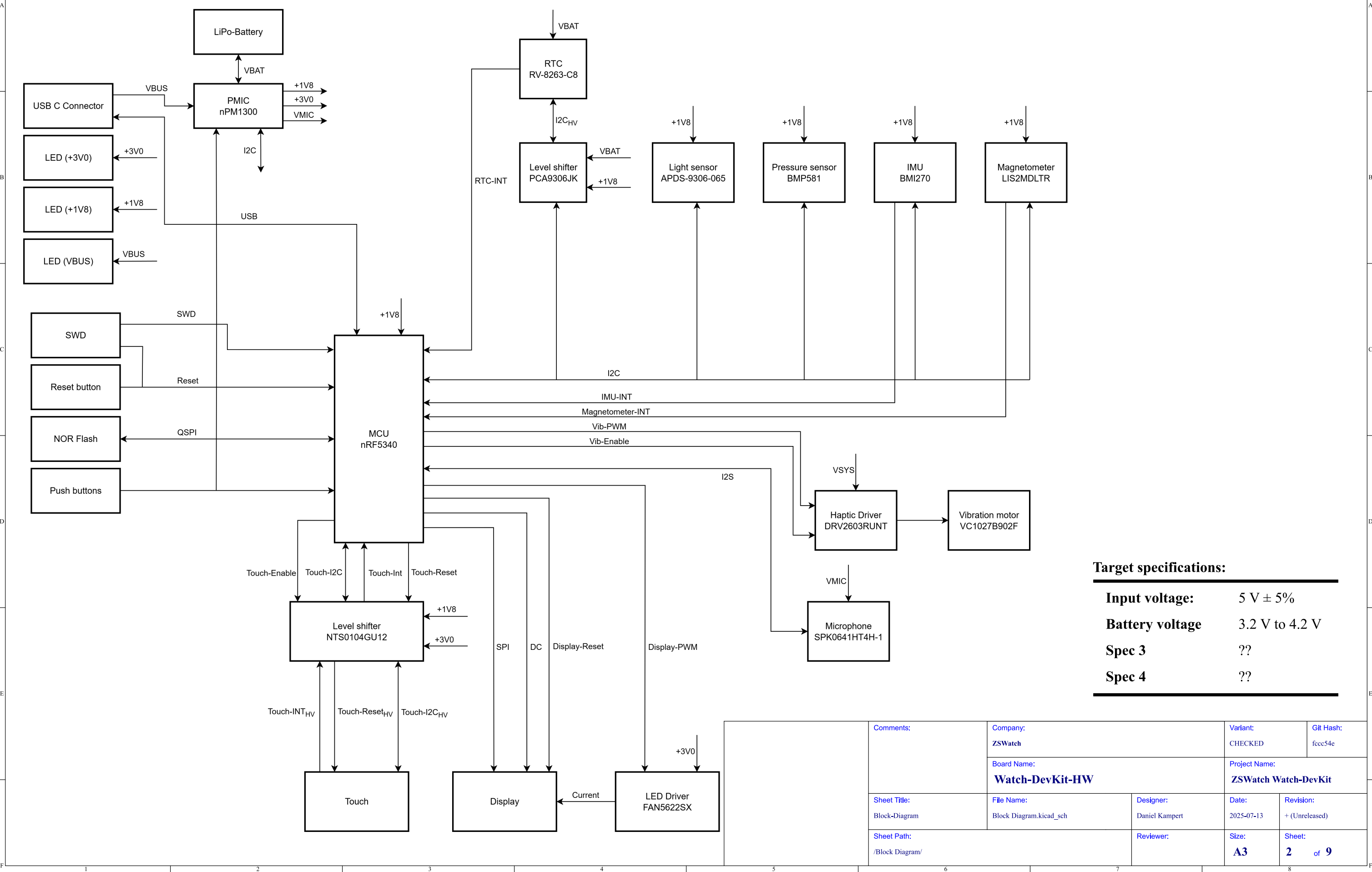
DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

LAYOUT NOTE:
Example text for
critical layout
guidelines.

	Comments:	Company: ZSWatch	Variant: CHECKED	Git Hash: fccc54e
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: ZSWatch-Watch	File Name: ZSWatch-Watch.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13
	Sheet Path: /		Reviewer:	Revision: + (Unreleased)
			Size: A3	Sheet: 1 of 9

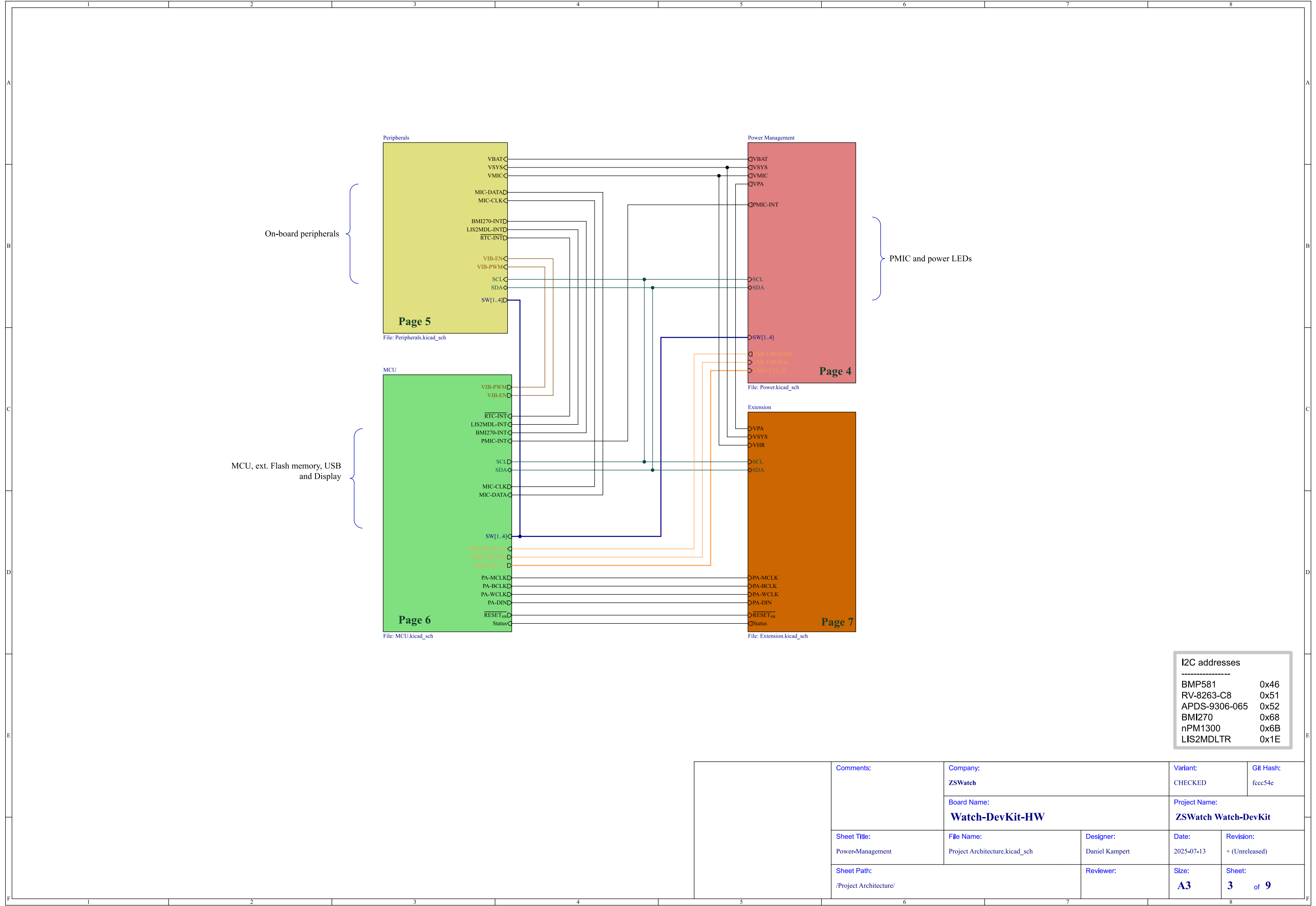
[2] Block-Diagram



	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

	Comments:	Company: ZSWatch		Variant: CHECKED	Git Hash: fccc54e
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title:	File Name: Power - Topology.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path: /Power - Topology/		Reviewer:	Size: A3	Sheet: 2 of 9



Peripherals

Power Management

MCU

Extension

Page 5

Page 4

Page 6

Page 7

File: Peripherals.kicad_sch

File: Power.kicad_sch

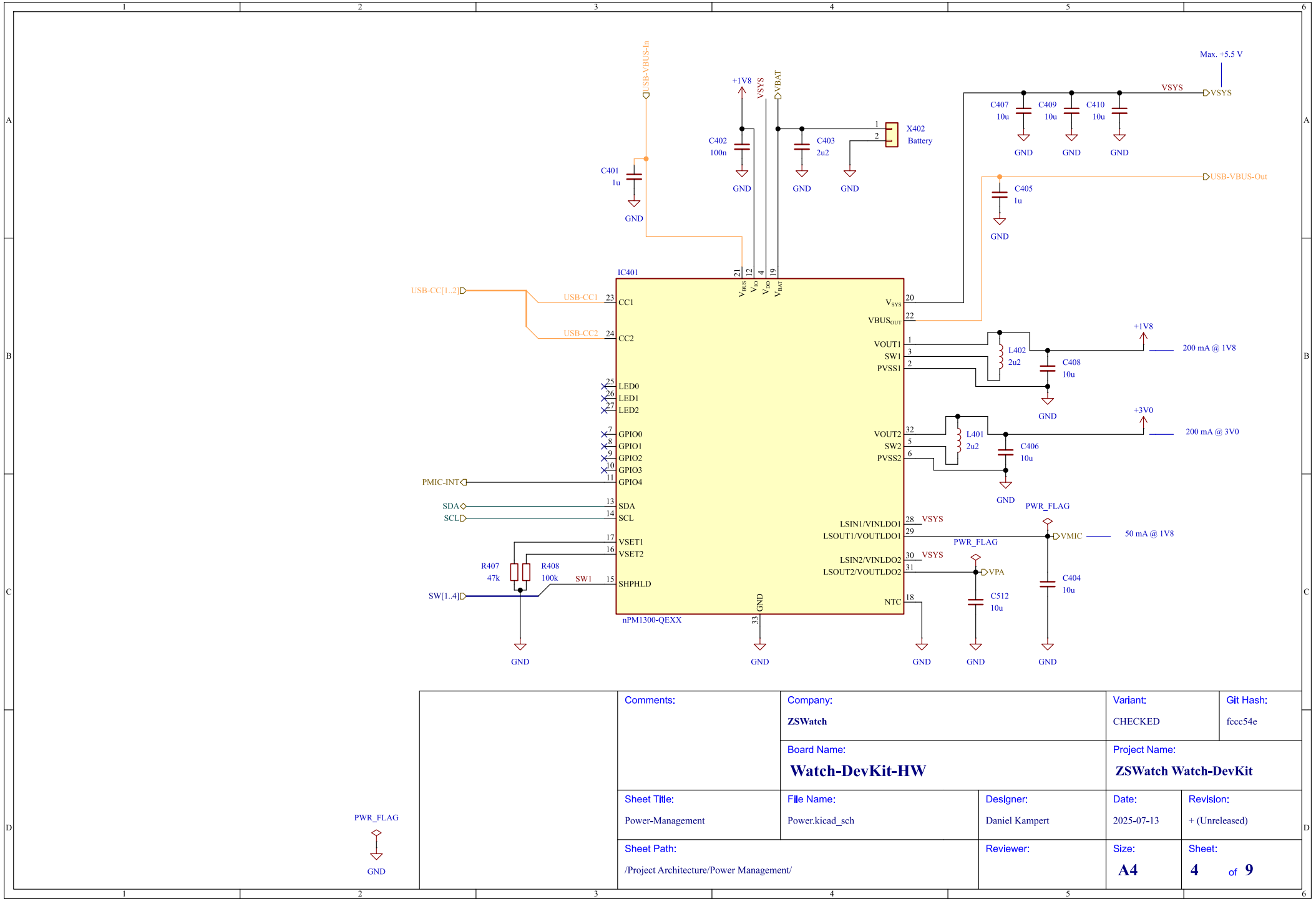
File: MCU.kicad_sch

File: Extension.kicad_sch

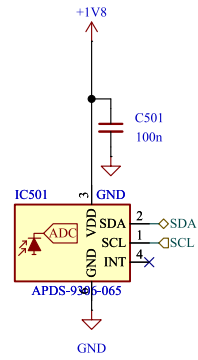
On-board peripherals

MCU, ext. Flash memory, USB and Display

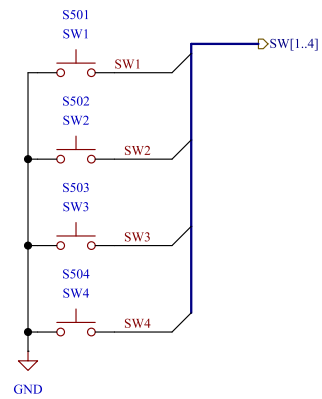
PMIC and power LEDs



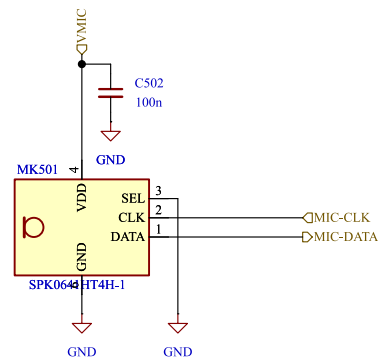
Light sensor



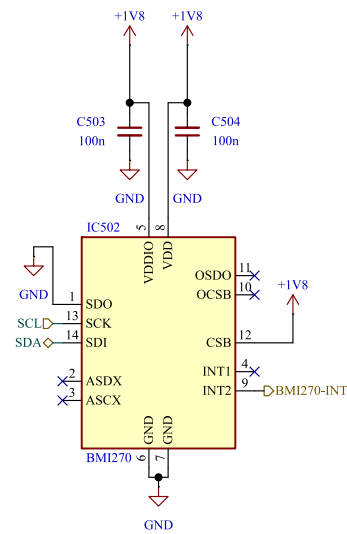
Buttons



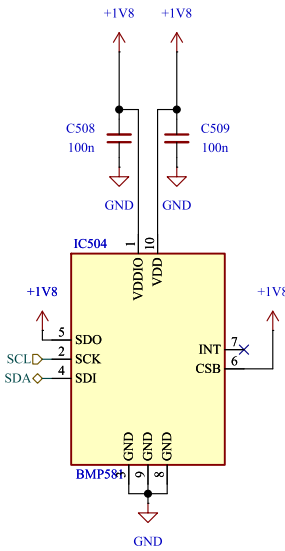
Microphone



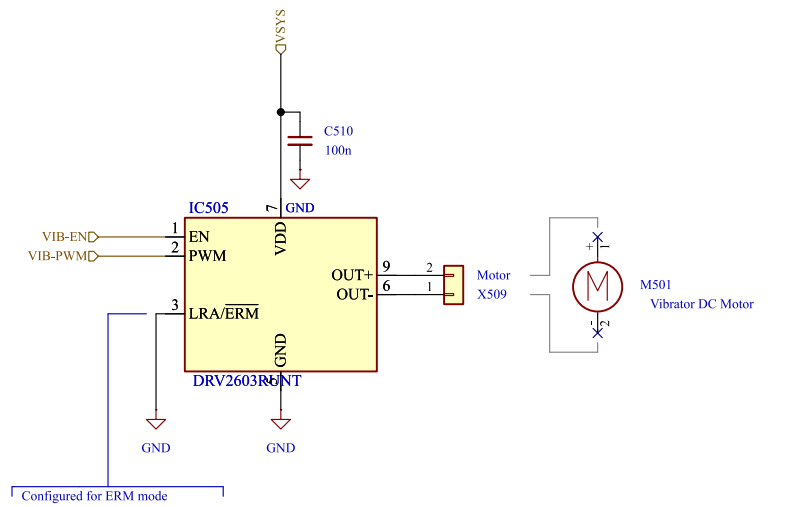
IMU with step counter and gesture detection



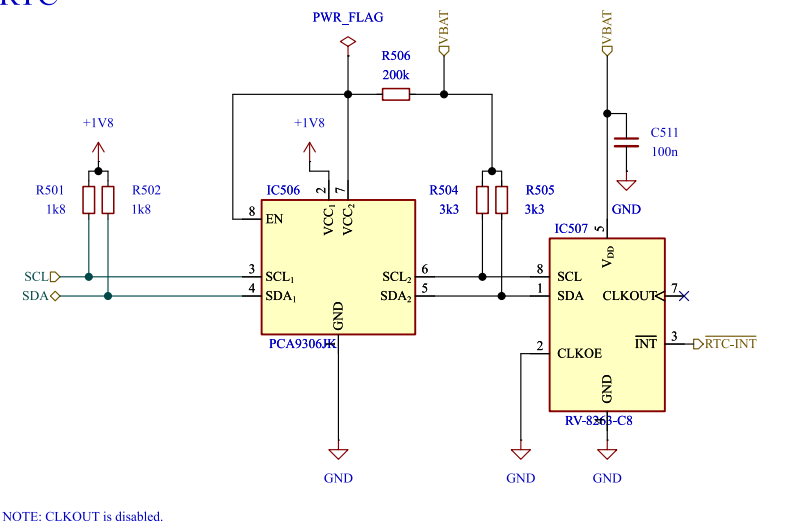
High accuracy pressure sensor



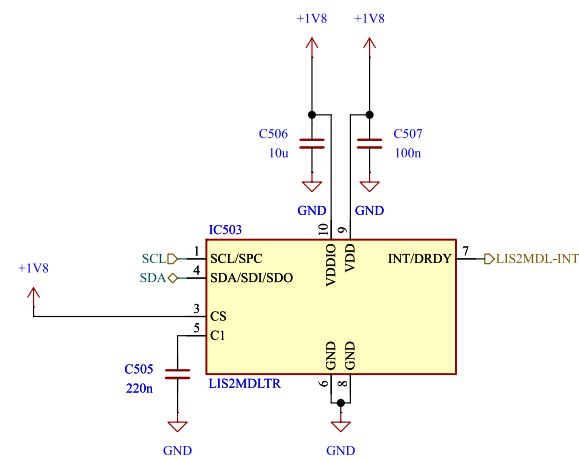
Vibration motor



RTC

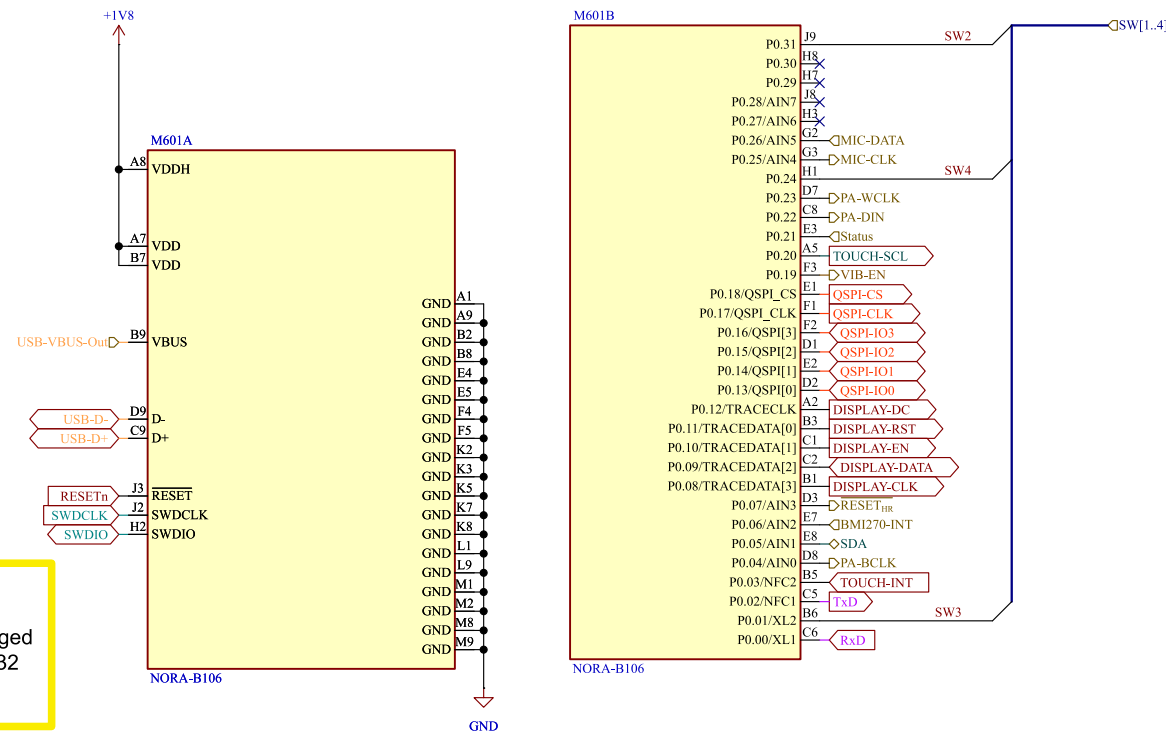


Magnetometer



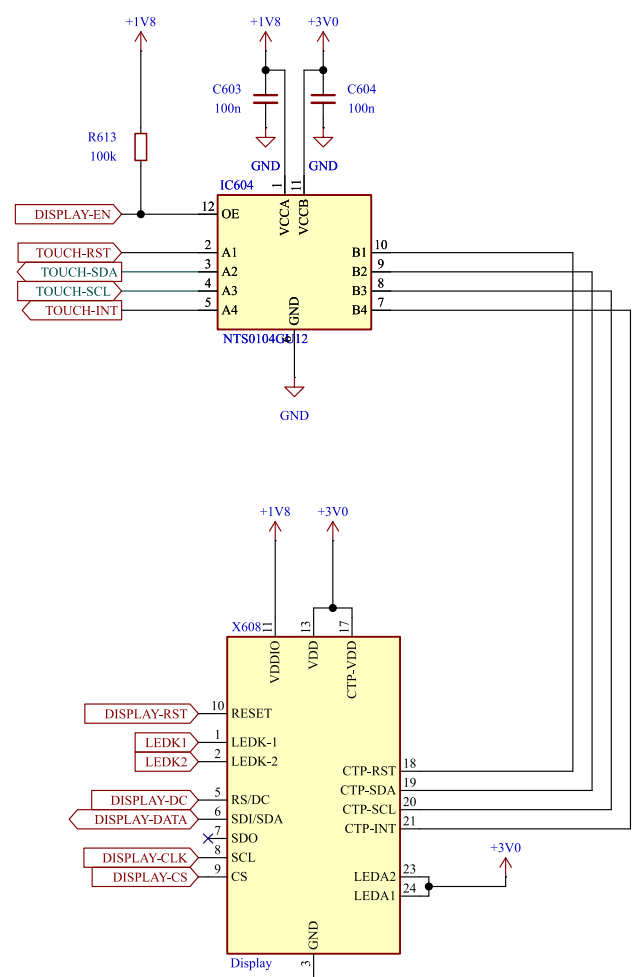
Comments:	Company: ZSWatch		Variant: CHECKED	Git Hash: fcc54e
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
Sheet Title: Peripherals	File Name: Peripherals.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Peripherals/		Reviewer:	Size: A3	Sheet: 5 of 9

Microcontroller
(NORA-B106/nRF5340)



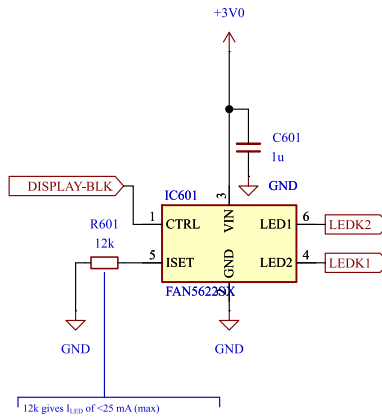
DESIGN NOTE:
The SPI lines for the display can't be changed and are required for 32 MHz operation.

Display

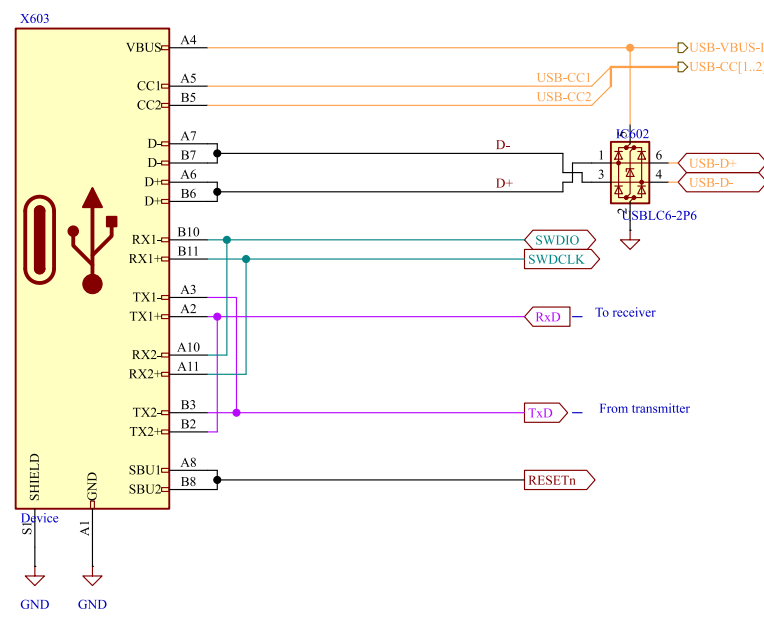


DESIGN NOTE:
NTS0104GU12 has internal pullups.

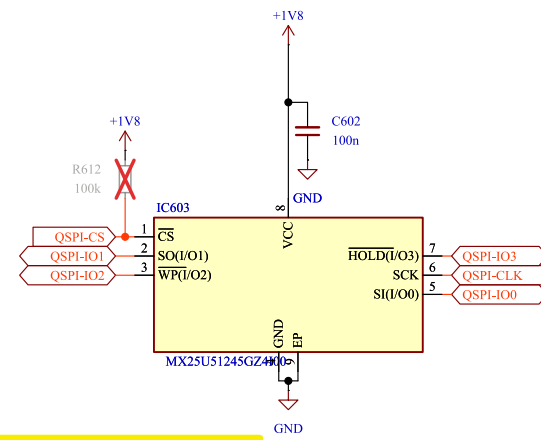
Display backlight



USB

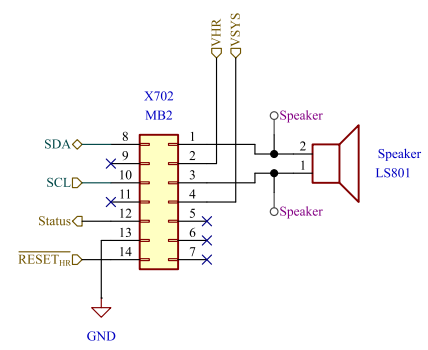
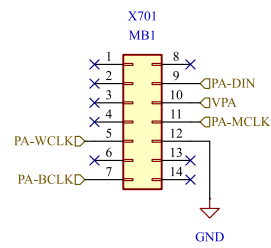


External Flash



DESIGN NOTE:
Make sure that the Flash Memory supports 4k sector erase commands (limited by QSPI driver)!

Comments: Sheet Title: MCU Sheet Path: /Project Architecture/MCU/	Company: ZSWatch		Variant: CHECKED		Git Hash: fcc54e
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit		
	File Name: MCU.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)	
		Reviewer:	Size: A3	Sheet: 6 of 9	



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		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Extension	File Name: Extension.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/Extension/		Reviewer:	Size: A4	Sheet: 7 of 9

[8] Revision-History

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	Comments:	Company: ZSWatch		Variant: CHECKED	Git Hash: fccc54e
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Revision-History	File Name: Revision History.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 9